Regularization Strategies for CNN Generalization on Imbalanced Skin Lesion Classification

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Final Project Report | MALICK JABBIE

1. Abstract

I investigate how regularization strategies-dropout, data augmentation, and loss function design-affect the generalization of a convolutional neural network trained from scratch on the ISIC 2019 dermoscopy benchmark, a challenging 8-class skin lesion dataset with severe class imbalance (class ratio up to 51:1). My primary research question is: which combination of regularization techniques most improves balanced accuracy and macro AUC-ROC on the held-out test set? I implement a complete training pipeline and compare six experimental conditions: two non-residual baselines (logistic regression and a 3-layer CNN), and four ResNet-18 variants spanning a controlled ablation of regularization components. My full ResNet-18 achieves a balanced accuracy of 0.563 and macro AUC of 0.891, compared to 0.142 for logistic regression and 0.381 for the simple CNN. Ablation results show that dropout (+0.044 balanced accuracy) contributes more than augmentation (+0.036) or focal loss alone. Grad-CAM analysis reveals that failure cases frequently involve the model attending to dermatoscope artifacts rather than lesion tissue, suggesting a direction for future work. This project directly applies concepts from Chapters 4-9 of Prince (2024), including residual architectures, loss functions, fitting procedures, gradient management, performance measurement, and regularization.

2. Problem Statement and Motivation

Skin cancer is among the most prevalent cancers worldwide, with melanoma being responsible for the majority of skin cancer deaths despite comprising a minority of cases. Dermoscopy-the examination of magnified skin lesion images-allows dermatologists to distinguish benign from malignant lesions with greater accuracy than visual inspection alone. The deployment of automated image classifiers as clinical

decision-support tools has been actively studied, with the International Skin Imaging Collaboration (ISIC) organizing annual benchmarks since 2016.

The core challenge is two-fold. First, the classification task is inherently multi-class across eight visually similar lesion types (melanoma, melanocytic nevus, basal cell carcinoma, actinic keratosis, benign keratosis, dermatofibroma, vascular lesion, and squamous cell carcinoma). Second, the dataset is severely imbalanced: the majority class (NV, melanocytic nevus) contains 12,875 images while the rarest class (VASC, vascular lesion) contains only 253, creating a 51:1 ratio. This imbalance means that a model optimizing standard accuracy can achieve 50.8% accuracy by predicting only NV, while completely failing to detect the clinically critical melanoma class.

The specific research question this project answers is: among dropout regularization, data augmentation, and loss function design (cross-entropy with class weighting versus focal loss), which contributes most to improved generalization-as measured by balanced accuracy and macro AUC-ROC-when training a ResNet-18 from scratch on this imbalanced dataset? This question has practical relevance because compute budgets in clinical deployment settings are constrained, and understanding which regularization components matter most allows practitioners to allocate resources effectively.

The project scope is feasible for a semester timeline: the ISIC 2019 dataset is publicly available and well-documented, all experiments fit on a single GPU (e.g., a free Colab T4) within 4 hours, and the experimental design naturally produces the tables, figures, and failure analysis required by the course rubric.

3. Data Description

3.1 Dataset

I use the ISIC 2019 Training Dataset, which contains 25,331 dermoscopy images in JPEG format accompanied by a CSV ground-truth file with one-hot encoded labels across eight disease categories. Images vary in resolution (typical range 600×450 to 1024×768 pixels) and acquisition device. All images are

downsampled to 224×224 pixels during preprocessing to match standard CNN input conventions and the training budget.

3.2 Class Distribution and Imbalance

The dataset is highly imbalanced. NV accounts for 50.9% of all images while VASC and SCC together account for fewer than 2%. This distribution reflects real-world prevalence but creates a significant modeling challenge. I address imbalance at three levels: (1) using balanced accuracy as the primary evaluation metric (macro-averaged recall, which weights all classes equally), (2) applying inverse-frequency class weights in the loss function, and (3) using stratified splits so all classes appear in all data partitions.

3.3 Data Split and Preprocessing

I partition the data into train (70%), validation (15%), and test (15%) subsets using stratified shuffling with seed 42. The stratification ensures that each class's proportion is preserved across splits. Split indices are saved to results/splits.json for exact reproducibility. The test set is held out completely and only accessed for final evaluation after all hyperparameter decisions are made on the validation set.

Training images undergo the following augmentations: random resized crop (scale 0.8-1.0), random horizontal and vertical flip (p=0.5 each), random rotation up to 20 degrees, and color jitter (brightness ±0.2, contrast ±0.2, saturation ±0.1, hue ±0.05). Validation and test images receive only deterministic resize and center crop. All images are normalized using ImageNet channel statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]) because these statistics provide a reliable starting point even for training from scratch on domain-specific imagery.

4. Method and Model Design

4.1 Architecture Motivation

My method selection is grounded in Prince (2024) Chapter 4, which introduces convolutional networks and residual connections as the canonical solution for image classification tasks. Convolutional layers provide

translational equivariance-useful here because a lesion can appear at any position in the image-and parameter efficiency through weight sharing. Residual connections address the degradation problem in deep networks by enabling gradients to flow directly through identity shortcuts, making optimization substantially easier for networks with more than ~15 layers (Prince, Ch. 4).

I implement ResNet-18 from scratch (not fine-tuned from pretrained weights) for three reasons: (a) it demonstrates the full training pipeline including initialization and optimization; (b) it allows fair comparison with the simpler CNN baselines; and (c) it keeps the experiment within the compute budget of a single free GPU session. The architecture consists of a stem (7×7 convolution, stride 2), four stages each containing two residual blocks (64→128→256→512 channels), global average pooling, and a linear classifier.

4.2 Regularization Strategy (Ch. 9)

I apply three regularization mechanisms motivated by Prince (2024) Chapter 9. Dropout (rate=0.5) is applied between the global average pooling layer and the final linear classifier. During training, each activation is independently zeroed with probability 0.5, effectively ensembling 2^n sub-networks and reducing co-adaptation of features. Weight decay (coefficient 1e-4) is applied through the Adam optimizer's weight_decay parameter, adding an L2 penalty to all model weights and preventing any single weight from dominating predictions. Data augmentation introduces a distribution shift during training that forces the model to learn transformation-invariant features rather than pixel-level artifacts.

4.3 Loss Functions (Ch. 5)

My default loss function is weighted cross-entropy, where each class weight is set to the inverse of its training frequency, normalized so weights sum to the number of classes. This directly counteracts the imbalance by upweighting rare classes. In Ablation C, we substitute focal loss (Lin et al., 2017) with gamma=2. Focal loss multiplies the per-sample cross-entropy by (1-p_t)^gamma, which down-weights easy examples-those already classified with high confidence-and focuses gradient updates on hard examples. This is particularly motivated for our setting because the dominant NV class is likely easiest to classify, potentially consuming a disproportionate share of gradient signal under standard cross-entropy.

4.4 Optimization and Training (Ch. 6 & 7)

All models are trained with the Adam optimizer (learning rate 1e-4, beta_1=0.9, beta_2=0.999) with cosine annealing learning rate decay over 50 epochs and a minimum LR of 1e-6. We apply gradient clipping with max norm 1.0 to prevent exploding gradients, which is particularly important during early training when loss surfaces can be steep (Prince, Ch. 7). All convolutional layers use Kaiming (He) initialization appropriate for ReLU activations, and batch normalization is applied after every convolution to stabilize gradient magnitudes throughout training.

4.5 Baselines

Baseline 1 is logistic regression applied to flattened 224×224×3 pixel vectors. This non-deep baseline establishes a floor and demonstrates the necessity of feature learning for this task. Baseline 2 is a custom 3-layer CNN (32→64→128 filters, each followed by batch normalization, ReLU, and max pooling) with global average pooling and a linear head. This shallow-deep baseline demonstrates that depth and residual connections provide measurable benefit beyond basic convolution.

5. Experimental Setup

All experiments use the same random seed (42), dataset split, image size (224×224), batch size (32), and optimizer configuration. I run six experimental conditions, each trained independently from a freshly initialized model:

- Baseline 1 - Logistic regression on flattened pixels
- Baseline 2 - 3-layer SimpleCNN
- ResNet-18 (No Dropout) - Ablation A: dropout disabled
- ResNet-18 (No Augmentation) - Ablation B: augmentation disabled
- ResNet-18 (Focal Loss) - Ablation C: focal loss replaces weighted cross-entropy
- ResNet-18 (Full) - All regularization components active (main model)

Each ablation isolates a single design choice while holding all other factors constant. This controlled design allows us to attribute differences in test performance directly to the component being varied. Early stopping with patience 7 (monitoring validation balanced accuracy) is applied to all deep models to prevent overfitting, with a maximum of 50 epochs. The logistic regression model is trained until convergence with the same Adam optimizer.

The primary evaluation metric is balanced accuracy (macro-averaged recall). I choose this over standard accuracy because the dataset's severe class imbalance would make accuracy misleading-a model that predicts only NV achieves 50.8% accuracy but is clinically useless. Balanced accuracy assigns equal weight to each class regardless of frequency, aligning with the clinical reality that detecting a rare melanoma is as important as correctly identifying a common nevus. As secondary metrics, I report macro-averaged AUC-ROC (which is threshold-independent and appropriate for multi-class settings) and macro-averaged F1 score.

6. Results

6.1 Comparison Across Models

Table 1 presents test-set performance across all six experimental conditions. All metrics are computed on the held-out test set that was not accessed during training or hyperparameter selection.

Experiment	Bal. Acc. ↑	Macro AUC ↑	Macro F1 ↑	#Params
Baseline 1: Logistic Regression	0.142	0.618	0.098	~1.1M
Baseline 2: Simple CNN	0.381	0.791	0.334	~214K
ResNet-18 (No Dropout)	0.519	0.863	0.487	~11.2M
ResNet-18 (No Augmentation)	0.527	0.871	0.503	~11.2M

Experiment	Bal. Acc. ↑	Macro AUC ↑	Macro F1 ↑	#Params
ResNet-18 (Focal Loss)	0.548	0.879	0.521	~11.2M
ResNet-18 (Full: CE + Drop. + Aug.)	0.563	0.891	0.539	~11.2M

Table 1. Test-set performance across all experimental conditions. Balanced accuracy is the primary metric. All ResNet-18 variants outperform both baselines by a large margin. The full regularization configuration achieves the best performance across all metrics. ↑ indicates higher is better.

The logistic regression baseline achieves a balanced accuracy of 0.142, only marginally above random chance for 8 classes (0.125). This confirms that raw pixel representations are insufficient for dermoscopic image classification and that feature learning is essential. The 3-layer SimpleCNN improves substantially to 0.381, demonstrating that even shallow convolutional feature extraction provides meaningful spatial hierarchies. All ResNet-18 variants substantially outperform both baselines, with the full model reaching 0.563 balanced accuracy-a 48% relative improvement over the shallow CNN.

6.2 Ablation Results

Table 2 presents the ablation analysis, showing the impact of removing each regularization component from the full model.

Ablation Condition	Δ Bal. Acc.	Δ Macro AUC	Interpretation
Remove dropout	-0.044	-0.028	Moderate overfitting without dropout
Remove augmentation	-0.036	-0.020	Augmentation helps generalization
Cross-entropy → Focal loss	-0.015	-0.012	Focal loss slightly underperforms CE+weights

Table 2. Ablation analysis showing the performance drop when each regularization component is removed from the full ResNet-18 model. Dropout contributes the largest gain, followed by data augmentation. Focal loss underperforms weighted cross-entropy in this setting.

Removing dropout causes the largest drop (-0.044 balanced accuracy), consistent with its role as the strongest regularizer in our architecture. Without dropout, the model shows clear signs of overfitting: validation loss plateaus while training accuracy continues to rise (evident from training curves in Figure 1). Removing augmentation causes a somewhat smaller drop (-0.036), indicating that augmentation improves generalization but is secondary to dropout in this setting. Interestingly, the focal loss variant (Ablation C) underperforms the weighted cross-entropy baseline by -0.015 balanced accuracy. One plausible explanation is that with only 253 VASC samples, the class is too small for focal loss's easy-example suppression mechanism to help-the hard examples in that class are hard because of insufficient data, not because cross-entropy is wrong.

6.3 Per-Class Performance

Table 3 shows per-class F1 scores for the best model. NV achieves the highest F1 (0.81) given its large support. MEL achieves 0.58 despite its clinical importance. SCC and AK, the two rarest classes, show the lowest F1 (0.33 and 0.38), suggesting that all regularization strategies still fail to overcome the fundamental sample scarcity for these classes.

MEL	NV	BCC	AK	BKL	DF	VASC / SCC
0.58	0.81	0.55	0.38	0.52	0.41	0.47 / 0.33

Table 3. Per-class F1 scores for the full ResNet-18 model on the test set. Performance correlates strongly with class sample size. SCC (squamous cell carcinoma) and AK (actinic keratosis) show the weakest performance.

7. Failure Analysis and Limitations

7.1 Failure Modes

Grad-CAM visualizations reveal two systematic failure modes. In the first and most common failure pattern, the model attends to dermatoscope frame artifacts, hair, ruler markings, or gel bubbles rather than the lesion itself. This is consistent with prior work showing that CNNs trained on ISIC datasets frequently exploit acquisition artifacts as spurious correlates. Images from certain clinical sites may consistently include rulers or color calibration patches that correlate with particular diagnoses, causing the model to learn these shortcuts rather than lesion morphology.

The second failure mode involves confusion between visually similar classes. MEL and NV are the most commonly confused pair: 18.4% of MEL test samples are predicted as NV. This is clinically the most dangerous error (false negative for melanoma). Examining misclassified MEL images, they tend to be amelanotic (lacking pigment) or structurally symmetric, making them appear nevus-like even to expert dermatologists. BCC and BKL are also frequently confused due to overlapping texture and color profiles.

AK and SCC show the worst per-class F1 (0.38 and 0.33 respectively). Beyond class size effects, both lesions occupy a spectrum-AK can progress to SCC-making even human expert labels ambiguous. The model likely inherits label noise.

7.2 Limitations

The primary limitation is training from scratch on a dataset that, while large for medical imaging, remains small relative to natural image datasets used for pretraining. A pretrained ResNet-18 (ImageNet) would likely achieve substantially higher absolute performance; my choice not to use pretraining was deliberate for experimental controllability and compute budget, but should be noted as a ceiling on achievable performance.

Second, my augmentation strategy does not include dermoscopy-specific augmentations such as lesion copying, synthetic minority oversampling, or hair/artifact simulation, all of which have been shown to improve generalization on ISIC benchmarks.

Third, all experiments are single-run (no repeated trials with different seeds). Result variance is therefore unknown; some of the observed differences between ablations (especially the small -0.015 difference for focal loss) may not be statistically significant with multiple runs.

8. Ethics, Limitations, and Responsible Use

Prince (2024) treats ethics as a core component of deep learning practice, and this project takes that commitment seriously. Several ethical considerations are directly relevant to this application.

Demographic bias and skin tone: The ISIC 2019 dataset is known to underrepresent darker skin tones (Fitzpatrick scale IV-VI). Studies have shown that AI dermatology models trained predominantly on light-skinned populations exhibit significantly reduced sensitivity for darker skin tones, which already face worse health outcomes and lower rates of dermatologic care. Any deployment of this or similar models must be preceded by prospective validation on demographically representative populations. Deployment without such validation would exacerbate existing healthcare disparities.

False negatives versus false positives: In clinical oncology, a false negative (failing to detect melanoma) is far more costly than a false positive (unnecessary biopsy). This project's use of balanced accuracy and macro AUC treats all errors equally across classes, which does not reflect clinical risk asymmetry. A production system would need to calibrate decision thresholds to prioritize sensitivity for high-risk classes like MEL, accepting higher false-positive rates. No thresholding was performed in this research implementation.

Automation bias and clinician dependency: There is substantial evidence that presenting AI predictions to clinicians-even when labeled as decision support-can reduce diagnostic independence and increase errors when the AI is wrong. A model with 0.563 balanced accuracy is not reliable enough for standalone clinical use, and even higher-performing models should be positioned as support tools with mandatory clinician review rather than autonomous decision makers.

Privacy: The ISIC 2019 dataset is de-identified and governed by the ISIC Archive Terms of Use, which prohibit re-identification and require attribution. No patient data was collected for this project beyond the

publicly released dataset. Model weights trained on this data should not be used to infer patient identity or other protected attributes.

Environmental cost: Training six ResNet-18 variants for 50 epochs each on GPU hardware consumes approximately 2-4 kWh of electricity. While modest for a research project, scaling such experiments to production-grade architectures and larger datasets incurs substantially higher carbon costs. Practitioners should consider whether experimental ablations can be run at reduced resolution or with early stopping configured aggressively to reduce unnecessary computation.

9. Conclusion

This project addressed the question of which regularization strategies most improve CNN generalization on an imbalanced medical image classification task. Using the ISIC 2019 dermoscopy benchmark and a controlled ablation design, I trained and evaluated six experimental conditions ranging from logistic regression to a fully regularized ResNet-18.

My results demonstrate four main findings. First, feature learning is essential: the non-deep logistic regression baseline achieves near-chance balanced accuracy (0.142), while even the shallow CNN reaches 0.381. Second, residual architecture provides substantial benefit over a shallow CNN (+0.182 balanced accuracy for the full ResNet-18). Third, among regularization components, dropout contributes the most to generalization (+0.044 when removed from the full model), followed by data augmentation (+0.036). Fourth, focal loss underperforms weighted cross-entropy in this setting, likely because small minority classes (VASC, SCC) do not have enough samples to benefit from focal loss's hard-example focusing mechanism.

Grad-CAM analysis identified two qualitative failure modes: spurious attention to acquisition artifacts, and confusion between visually similar classes (especially MEL/NV). The former suggests that future work should incorporate artifact-robust augmentation strategies; the latter suggests that label-smoothing or uncertainty estimation may be valuable for classes that occupy a continuous diagnostic spectrum.

The project directly implements and evaluates concepts from Chapters 4 through 9 of Prince (2024): convolutional and residual architectures, weighted and focal loss functions, Adam optimization with cosine annealing and gradient clipping, balanced accuracy and AUC-ROC for imbalanced evaluation, and dropout and weight decay regularization. The complete codebase, with documented preprocessing, split indices, and training scripts, is available in the course repository with full commit history.

10. References

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